Development of A Brain Tumor Detection System using Convolutional Neural Network

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ORIGINAL RESEARCH

Abstract—Brain tumor is a cancerous or noncancerous growth of abnormal cells in the brain. These growths cause severe health complications due to the limited space within the skull, often leading to headaches, loss of physical abilities, and personality disorders. Early and accurate detection is therefore critical. In this research, we developed a brain tumor detection system based on Convolutional Neural Networks (CNNs). A total of 371 MRI images from the Brain Tumor Segmentation (BraTS) dataset were used to train and evaluate three architectures: Fully Convolutional Network (FCN), Region-based CNN (R-CNN), and YOLO. The models were compared in terms of detection performance, with R-CNN achieving the highest accuracy of 99.45%, followed by FCN at 98.43% and YOLO at 97.82%. The proposed system outputs not only the presence of a tumor but also highlights its location, size, and structure within the MRI scans. These findings demonstrate the potential of CNN-based systems in improving brain tumor detection for clinical decision support.

Keywords— Accuracy, Brain Tumor, Convolutional Network, Detection, Segmentation

1 INTRODUCTION

A brain tumor is the growth of abnormal cells in the brain. Originating in the glial cells, gliomas are the most common brain tumor (Ferlay *et. al.*, 2010). The location of the tumor in the brain creates problems such as loss of some physical abilities, headache, nausea, vomiting, seizures, vision issues and personality disorders. Brain tumor can be grouped into two main types which are Benign (Noncancerous) and Malignant (Cancerous) tumors. Benign (Noncancerous) tumor is the one that does not grow suddenly and has no effect on tissue and Malignant (Cancerous) tumor grows rapidly and affects the neighboring tissue and with time, it affects human life and leads to death.

Tumor detection is important in the treatment process and these tumors can be diagnosed using Computed Tomography (CT), x-rays and Magnetic Resonance Imaging (MRI). Every detail regarding the development of the human brain and the detection of abnormal cell growth in the brain can be detected by MRI (Harshit *et al.*, 2022).

Image segmentation is a very important stage in tumor detection, in which tumor is being separated from other healthy tissues and helps in changing the appearance of an image. By isolating the tumor then determines its stage, the treatment becomes easier (Marcel, 2004). In recent years, deep learning-based technique have achieved great success in the field of computer vision and medical imaging.

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2 RELATED WORKS

Ali *et al* (2021) presented a review of brain image segmentation in recent years. Several segmentation methods were discussed from simple intensity-based to high-level segmentation approaches such as metaheuristic, machine learning, deep learning and hybridization. Advantages and disadvantages of brain image segmentation methods are also discussed to give a better understanding of the strengths and weaknesses of the existing methods. Deep learning-based and hybrid-based metaheuristics approaches are found to be more efficient for reliable segmentation of brain tumors but these methods however fall behind in terms of computation and memory complexity.

Biratu *et al* (2021) proposed a survey of brain tumor segmentation and classification algorithms. A number of tasks have been performed in the area of brain tumor classification into their respective histological type and an impressive performance results have been obtained. The purpose of the paper provided a comprehensive survey of three, recently proposed, major brain tumor segmentation and classification model techniques which are region growing, shallow machine learning and deep learning considering the state-of-the-art methods and their performance. The established works included in this survey also covers technical aspects such as the strengths and weaknesses of different approaches, pre- and post-processing techniques, feature extraction, datasets, and models' performance evaluation metrics.

Isselmou et al. (2021) proposed a deep wavelet auto-encoder (DWAE) model to classify MRI brain images as tumor or non-tumor, achieving a very high accuracy of 99.3%. While the method demonstrated excellent performance, its reliance on handcrafted wavelet transformations limits its adaptability compared to end-to-end CNN-based models.

Similarly, Omodunbi et al. (2021) developed a hybrid machine learning model for diabetes prediction using Light Gradient Boosting Machine (LGBM) and K-Nearest Neighbor (KNN). Although not directly focused on brain tumors, their work shows how hybridization of models can enhance prediction accuracy. Traditional ML approaches like these often require extensive feature engineering and are less efficient when dealing with high-dimensional MRI data compared to CNNs.

Ramin et al. (2021) explored brain tumor segmentation using a Cascade CNN with a Distance-Wise Attention (DWA) mechanism, which improved tumor localization. Their approach highlights the importance of attention mechanisms in enhancing accuracy, but the framework was designed primarily for segmentation, not real-time detection, and may be computationally intensive for smaller datasets.

Adesh Kumar (2022) compared several traditional segmentation techniques, including Otsu's, watershed, level set, K-means, DWT, and CNN. Results showed CNN outperforming other methods with 91.39% accuracy, reinforcing the strength of deep learning for medical imaging. However, the study only compared algorithms for segmentation and did not extend into detection pipelines or real-world system development.

Harshit et al. (2022) proposed a CNN-based automated detection and segmentation framework using a Kaggle dataset, achieving 95.6% accuracy. While the study demonstrated the value of deep networks for glioma localization, it focused heavily on segmentation and classification separately, without benchmarking different CNN architectures under a unified detection framework.

Taken together, these works demonstrate the effectiveness of deep learning for brain tumor analysis, yet most studies emphasize segmentation or classification in isolation, often on large, curated datasets. Few have attempted to benchmark and compare detection-oriented CNN architectures such as R-CNN and YOLO alongside segmentation-oriented ones like FCN within the same system. This creates a clear research gap, which this study addresses by developing a unified CNN-based brain tumor detection system and evaluating its performance using multiple architectures on the BraTS dataset.

Veeramuthu *et al* (2022) proposed a combined feature and image-based classifier (CFIC) for brain tumor image classification. The dataset was gotten from Kaggle brain tumor detection 2020 and theses dataset were used to train and test the proposed classifiers. Different deep neural network and deep convolutional neural network-based architecture were proposed where we have Actual image feature-based classifier (AIFC), Segmented image-based classifier (SIC), Actual and segmented image-based classifier (ASIC) and Combined feature and image-based classifier (CFIC). CFIC performs better compared to the various classifiers proposed, it has a sensitivity of 98.86%, specificity of 97.14% and accuracy of 98.97%.

Mahmud *et al* (2023) presented a deep analysis of brain tumor detection from MR images using deep learning networks. With the aid of magnetic resonance imaging (MRI), allows for the quick and simple identification of brain tumors. A convolutional neural network (CNN) architecture was suggested in this study for the efficient identification of brain tumors using MR images. This paper also discussed various models such as ResNet-50, VGG16, and Inception V3 and conducted a comparison between the proposed architecture and these models. To analyze the performance of the models, different metrics were considered such as the accuracy, recall, loss and area under the curve (AUC). As a result of analyzing different models with the proposed model using these metrics, the conclusion is that the proposed model performed better than the others. A dataset of 3264 MR images was used, the CNN model had an accuracy of 93.3%, an AUC of 98.43%, a recall of 91.19%, and a loss of 0.25. Therefore, the proposed model is reliable for the early detection of a variety of brain tumors after comparing it to the other models.

Although numerous studies have applied Convolutional Neural Networks (CNNs) for brain tumor analysis, most existing works have primarily focused on segmentation or classification using single architectures and large, curated datasets. Comparative studies do exist, but they often lack consistency in evaluation frameworks and rarely address detection-focused architectures such as R-CNN and YOLO alongside segmentation-based networks like FCN. Furthermore, limited attention has been given to scenarios involving relatively small datasets, where model performance, adaptability, and robustness are crucial for real-world clinical applications. This creates a gap for research that not only compares diverse CNN models within a unified system but also demonstrates how such a system can be developed and optimized for practical brain tumor detection.

3 METHODOLOGY

I. Dataset and Data Description

This research on Brain Tumor Segmentation Using CNN, utilizes the Brain Tumor Segmentation (BraTS) multimodal scans dataset. This dataset is made up of multimodal Magnetic Resonance Imaging (MRI) scans available in the Neuroimaging Informatics Technology Initiative (NIfTI) format, which is a widely used medical imaging format. The scans were produced using diverse MRI settings and are composed of four types of images:

1. **T1-weighted images**: These are native images obtained through sagittal or axial 2D acquisitions, having a slice thickness of 1–6 mm.

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2. **T1ce-images**: These are T1-weighted images that have been enhanced using Gadolinium. They are collected using 3D acquisition and have a 1 mm isotropic voxel size for most patients.

3. **T2-weighted images**: These images are obtained using axial 2D acquisition, featuring a slice thickness of 2–6 mm.

4. **FLAIR images**: These are T2-weighted FLAIR images that have been acquired axially, coronally, or sagittally using 2D acquisitions, and they have a slice thickness of 2– 6 mm.

The gathered data was collected from multiple sources, specifically 19 different institutions, and therefore, comes with diverse clinical protocols and various scanner types. Moreover, all the imaging datasets have been manually segmented by one to four raters, all following the same annotation protocol. These raters' annotations have been further approved by experienced neuro-radiologists to ensure their accuracy.

The annotations consist of three types of labels:

1. **GD-enhancing tumor (ET — label 4)**,

2. **Peritumoral edema (ED — label 2)**,

3. **Necrotic and non-enhancing tumor core (NCR/NET — label 1)**.

The provided data have undergone pre-processing, which includes co-registering to the same anatomical template, interpolating to the same resolution of 1 mm^3, and skull-stripping.

A Data Pre-Processing

The main tumor subregions considered in this study are the necrotic core, edema, and enhancing tumor, as these provide essential insights into tumor morphology and progression. We obtained a total of 371 MRI images from the BraTS dataset, which were preprocessed and used for training and evaluation. Unlike the full BraTS dataset, which consists of hundreds of patients and multiple modalities, our subset contained 371 representative images that were selected and curated for this experiment. This subset size was chosen to balance tumor and non-tumor classes and to ensure computational feasibility for the models under study.To reduce noise and avoid uninformative slices with little or no brain tissue, the first and last 5 slices of each volume were discarded, leaving more clinically relevant data. Following domain practices (Menze et al., 2015; Bakas et al., 2018), the slices were further restricted to 100 per volume starting from the 22nd slice, which better captures the tumor's core and surrounding pathology. Each slice was resized to a standard input dimension of 256 × 256 pixels and normalized to the range [0,1] to account for intensity variations across scans.

B. Data Pre-Splitting

The dataset was divided into training (70%), validation (15%), and testing (15%) subsets. The training set enabled the model to learn tumor features, the validation set was used for hyperparameter tuning and to prevent overfitting, and the test set provided an unbiased performance evaluation. To enhance model generalization, data augmentation (random rotations, horizontal flips, and slight intensity shifts) was applied. A custom Data Generator class was implemented, inheriting from keras.utils.Sequence, which allowed batch-wise data preparation, efficient GPU utilization, and shuffling after each epoch to prevent bias. Ground-truth segmentation masks were converted into one-hot encoded arrays and resized to match input dimensions.

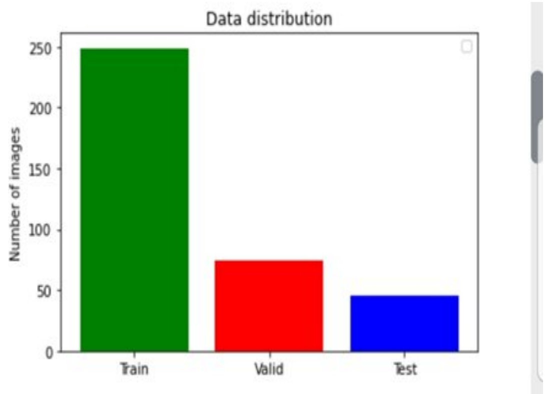

Figure 1: details of the data splitting process

C. Model Classification

All models were trained with the Adam optimizer (learning rate = 0.0001), a batch size of 16, and for up to 50 epochs, with early stopping applied to reduce overfitting. Experiments were conducted in Python (TensorFlow/Keras)

D. Model Implementation

All experiments were implemented in Python 3.9 using TensorFlow 2.9 with Keras backend. The images were resized to 256 × 256 pixels and normalized to a range of [0,1]. Data augmentation included random rotations (±15°), horizontal/vertical flips, and slight zooms (±10%) to improve generalization.

The models were trained using the Adam optimizer with a learning rate of 0.0001, a batch size of 16, and for up to 50 epochs. Categorical cross-entropy loss was used for training FCN and R-CNN, while binary cross-entropy loss was used for YOLO classification tasks. Early stopping was applied with a patience of 10 epochs based on validation loss.

The dataset was split into 70% training, 15% validation, and 15% testing, and data was shuffled after each epoch to minimize bias. Evaluation metrics included Accuracy, Precision, Recall, F1-score, Dice Similarity Coefficient, and Intersection over Union (IoU),

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E. Model Evaluation

This research work uses several evaluation metrics to assess the performance of our segmentation models. These metrics include:

Dice Coefficient:
The Dice coefficient is a common evaluation metric used in image segmentation tasks. It measures the similarity between the predicted segmentation mask and the ground truth mask. The formula for calculating the Dice coefficient is as follows:
Dice = (2 * Intersection) / (Sum of pixels in the predicted mask + Sum of pixels in the ground truth mask)
The Dice coefficient ranges from 0 to 1, where a value of 1 indicates a perfect match between the predicted and ground truth masks.

Dice Loss:
The Dice loss is the complement of the Dice coefficient. Instead of maximizing the Dice coefficient, the Dice loss was minimized during training. The Dice loss is calculated as (1 - Dice coefficient). It measures the dissimilarity between the predicted and ground truth masks. By minimizing the Dice loss, the model was encouraged to produce more accurate and precise segmentations.

$$DSC=|X|+|Y|2\times|X\cap Y|$$

Per Class Evaluation of Dice Coefficient:
In addition to the overall Dice coefficient, class-specific Dice coefficients were calculated for different tumor classes, namely necrotic, edema, and enhancing. These class-specific Dice coefficients provide insights into how well the model performs for each tumor class individually. Each class-specific Dice coefficient is calculated using the same formula as the overall Dice coefficient but applied to a specific class of interest.

Precision:
Precision is a metric that quantifies the proportion of correctly predicted positive samples out of all predicted positive samples. In this context, precision measures how well the model identifies tumor regions among the predicted positive regions. The precision metric helps evaluate the model's ability to minimize false positives.

Sensitivity:
Sensitivity, also known as recall or true positive rate, measures the proportion of correctly predicted positive samples out of all actual positive samples. In tumor segmentation, sensitivity indicates how well the model identifies tumor regions among the ground truth positive regions. A higher sensitivity indicates that the model is effective in capturing true positive tumor regions.

Specificity:
Specificity measures the proportion of correctly predicted negative samples out of all actual negative samples. Specificity indicates how well the model identifies non-tumor regions among the ground truth negative regions. Higher specificity values suggest that the model is good at correctly identifying regions without tumors.Figure2 shows theblock diagram of brain tumor detection model

Figure 2: Block diagram of brain tumor detection model

4 RESULT ANALYSIS

The three models FCN (Fully Convolutional Network), RCNN (Region-based Convolutional Neural Network), and YOLO (You Only Look Once) were evaluated on a test data. Below in Table 1 is the analysis of the results and comparison of the performance of these models on the test data:

Table 1: Test Results

	N	RCNN	YOLO
Loss	0.0859	0.0226	0.1356
Accuracy	0.9841	0.9919	0.9782
Mean Intersection Over Union (IOU)	0.3756	0.8284	0.6107
Dice Coefficient	0.2579	0.6369	0.2592
Precision	0.9840	0.9925	0.9785
Sensitivity	0.9840	0.9903	0.9780
Specificity	0.9947	0.9975	0.9928
Dice Coefficient for Necrotic Region	0.0109	0.5976	0.0095
Dice Coefficient for Edema	0.0551	0.7991	0.0562
Dice Coefficient for Enhancing Region	0.0218	0.6839	0.0148

Based on these results, the 1R-CNN model demonstrated the best overall performance across the metrics, followed by the FCN, while YOLO achieved the lowest scores. Although the performance values are high, they should be interpreted with caution. The relatively small dataset (371 cases) and slice-selection preprocessing steps may have reduced noise and contributed to inflated performance compared to reports in the wider literature. When compared to state-of-the-art works using larger BraTS datasets, which often report Dice scores in the range of 0.80–0.90 for brain tumor segmentation, the results here appear optimistic. This suggests the possibility of overfitting and limited generalizability.

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Therefore, these outcomes should be considered as preliminary and dataset-specific, rather than definitive evidence of superiority.

5 CONCLUSION

In this work, we applied three deep learning models, R-CNN, FCN, and YOLO, for brain tumor detection and segmentation using MRI images. The results we obtained look promising, but they should be treated with caution. Our dataset was relatively small (371 MRI volumes), and the preprocessing choices may have limited the diversity of the data. Because of this, the very high accuracy scores we reported are unlikely to hold up on larger or more varied clinical datasets. Another limitation is that we did not test the models on an independent dataset, which makes it hard to know how well they would perform in real-world medical practice. Looking ahead, there is plenty of room to strengthen this work. Larger, multi-institutional datasets with expert annotations would give a more reliable picture of model performance. Since getting such medical data is often difficult, techniques like transfer learning, federated learning, or semi-supervised annotation could be useful alternatives. It will also be important to use more robust validation strategies and evaluation metrics to make sure results are both fair and comparable to state-of-the-art studies.

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